

Sarvesh Karandikar

Software Engineer

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EXPERIENCE

Software Engineer — Nexus Dashboard

2023-11 – present

[Cisco Systems](#)

- Designed and implemented a Certificate Management System for automated root and intermediate CA creation, secure certificate signing, and integrity of Kubernetes-based communications.
- Enhanced external CA integration to securely onboard Cisco XE/XR devices, streamlining enterprise deployment and configuration.
- Developed an agent-based interface for Postgres and time-series DB management, enabling secure superuser access with application-specific flexibility.
- Implemented an SPI-based leader-election framework in Java for active-standby configurations with dynamic leader selection.

Software Engineer — Cloud Network Controller

2020-07 – 2023-10

[Cisco Systems](#)

- Integrated Google Cloud Platform into the ACI model-based cloud networking solution, enabling multi-cloud compatibility and scalability.
- Enhanced a configuration-drift platform to detect security-group rule drift across L4–L7 service device chains, improving compliance checks.
- Built a centralized Terraform-based collection service aggregating multi-cloud config into a unified graph database.
- Created a firewall network-configuration analysis service using Binary Decision Diagrams to assess connectivity and traffic flows.

Website Developer

2019-04 – 2020-05

[USC Global Supply Chain Management](#)

- Designed and continuously improved websites, including a redesign of a global supply-chain summit event site with sponsor management and full technical ownership.

Software Engineer Intern

2019-05 – 2019-08

[Circle Link](#)

- Implemented Named Entity Recognition to classify entities in loan documents, enabling extraction and Elasticsearch indexing.
- Fine-tuned an OCR-to-PDF model to recognize MICR formats on checks, streamlining loan tracking.

PROJECTS

DCGAN Image Synthesis — Deep Convolutional GAN generating high-quality images from noise, trained on CIFAR-10 with batch norm and adversarial training. [repo ↗](#)

Personalized FAQ Chatbot — Local FAQ chatbot using LLaMA 2-7B to vectorize PDFs for real-time, context-aware question answering on a 12GB GPU.

Satellite Road Detection — Classical computer-vision pipeline (morphology + K-means + edge detection) that highlights roads in satellite imagery. [repo ↗](#)

LSTM Text Generator — From-scratch NumPy RNN/LSTM that generates prose word-by-word; reaches ~90% next-word accuracy. [repo ↗](#)

CIFAR-10 Image Classifier — Convolutional neural network classifying CIFAR-10 images at ~75% test accuracy. [repo ↗](#)

AlpacaBot Trading Dashboard — Terminal-themed dashboard monitoring a weekly-options trading bot: positions, trades, and performance. [repo ↗](#)

WhatsApp AI Assistant — Production WhatsApp-native AI assistant on AWS Lightsail with per-user state, 17 slash-commands, and HMAC security. [repo ↗](#)

WhatsApp Media Bridge — Bridges WhatsApp to a Jellyfin/Sonarr/Radarr media stack with local Llama + Claude model routing. [repo ↗](#)

SKILLS

Languages: Go · Python · Java · C++ · GraphQL

Orchestration: Docker · Kubernetes · OCI · Stacker

Cloud: AWS · Azure · GCP

Frameworks: TensorFlow · PyTorch · Pandas · NumPy · NLTK · Terraform

Databases: Dgraph · Postgres

EDUCATION

University of Southern California — M.S. Computer Science (2020-05)

SRM Institute of Science and Technology — B.Tech Computer Science and Engineering (2018-05)